

Planetary Rover — Single Rover, Static Environment

AI4Ro2 · University of Genova · 2025-26
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Domain & Problem Description

A single wheeled rover navigates a static directed graph of locations. It starts at a base station and must collect geological samples from target sites and return them to base. Every move consumes battery. The environment is fully known in advance — no uncertainty, no dynamic obstacles. **Types:** location, sample. **Predicates:** at-rover, connected, sample-at, carrying, delivered, is-base. **Fluents:** battery-level, move-cost.

Modelling Choices

Q1 — Classical PDDL with numeric fluents. Battery is a global numeric fluent. The move action has a hard precondition ($\geq$ battery-level move-cost) and decreases it on every traversal. Three actions separate task phases: move, collect-sample, deliver-sample. This forces genuine energy reasoning — the planner cannot move when charge is insufficient.

Q2 — PDDL+ with continuous processes and safety event.

Construct	Name	Effect / Role
Process	idle-drain	Decrease battery-level (* #t idle-drain-rate) — always active
Process	move-drain	Decrease battery-level (* #t move-drain-rate) — only while rover-active
Event	battery-critical	Fires when battery $\leq$ threshold; sets (not system-ok) — shuts down all actions

Key insight: idle time has a real cost. Battery drains even while collecting. The critical event fires autonomously — the planner cannot control it, only race against it. Mission feasibility depends on completing all goals before the threshold is reached.

Plan Output

Q1-P1 (battery=60): 4-step plan — move, collect, move, deliver. Battery remaining: 40. Trivial, as expected.

Q1-P2 (battery=130): 18-step plan — 3 separate trips. Battery=130 is the exact minimum; any detour fails. Planner correctly routes via shortest paths. Matches expectations.

Q2-P1 (battery=80 OK / battery=60 FAILS): With 60, critical event fires at t=23 during return — the 5-unit idle drain during collection is the deciding factor. With 80, plan completes at t=26 with battery=14 above threshold. Q1 model would have declared 60 sufficient — this is the key difference.

Q2-P2 (battery=310): Two trips via waypoint, 84 time units total. Battery reaches exactly 10 at delivery (threshold=5). Trip ordering is critical — fetching the farther sample first would exhaust battery before return.

Limitations

No geometry — locations are symbolic; terrain, gradient, and slip are not modelled.

No uncertainty — real rovers face sensor noise, hardware faults, and comm delays.

No recharge — solar/RTG charging absent; battery is strictly decreasing.

Task-motion gap — collect-sample hides instrument deployment (itself a planning problem).

Linear drain only — real degradation is non-linear (temperature, battery age).